Evaluate

$$\int_{-\infty}^{\infty} e^{-\pi x^2} \cdot e^{-2\pi i x \xi}\, dx = e^{-\pi \xi^2} \quad (\xi > 0)$$

Proof. Let $f(z) = e^{-\pi z^2}$. Then, it is entire.

Consider four contour : $C_1 : t \ (-R \le t \le R)$,

$C_2 : R + it \ (0 \le t \le \xi)$

$C_3 : t + i\xi \ (-R \le t \le R)$

$C_4 : -R + it \ (0 \le t \le \xi)$

Then, the composition $C = C_1 \cdot C_2 \cdot (-C_3) \cdot (-C_4)$ is described as



,

and, by the Cauchy's theorem, $\int_C f = 0$. Now,

$$\int_{C_1} f = \int_{-R}^{R} e^{-\pi x^2}\, dx,$$

$$\int_{C_2} f = \int_0^{\xi} e^{-\pi (R+it)^2} \cdot i\, dt = i \cdot \int_0^{\xi} e^{-\pi (R^2 - t^2 + 2iRt)} \cdot dt$$

and so $\left| \int_{C_2} f \right| = \left| \int_0^{\xi} e^{\pi \cdot (t^2 - R^2)}\, dt \right| \le \left| \int_0^{\xi} e^{\pi t^2}\, dt \right| \cdot e^{-\pi R^2} \to 0$ as $R \to \infty$.

Similarly $\left| \int_{C_4} f \right| \to 0$ as $R \to \infty$. And,

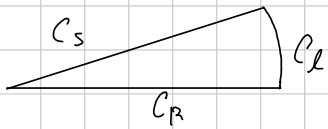
$$\int_{C_3} f = \int_{-R}^{R} e^{-\pi (t + i\xi)^2}\, dt = \int_{-R}^{R} e^{-\pi (t^2 - \xi^2 + 2it\xi)}\, dt$$

$$= e^{\pi \xi^2} \cdot \int_{-R}^{R} e^{-\pi t^2} \cdot e^{-2\pi i t \xi}\, dt$$

Thus, as $R \to \infty$,

$$0 = \int_{-\infty}^{\infty} e^{-\pi x^2}\, dx - e^{\pi \xi^2} \cdot \int_{-\infty}^{\infty} e^{-\pi t^2} \cdot e^{-2\pi i t \xi}\, dt. \quad \square$$

For each $n=2,3\cdots$ evaluate $\int_0^\infty \frac{1}{1+x^n}\,dx$.



Proof. Let $f(z)=\frac{1}{1+z^n}$. Take a contour C as a composition of:

$C_R: t\mapsto t \quad (0\le t\le R)$, $C_\ell: t\mapsto R\cdot e^{\frac{it}{n}} \quad (0\le t\le 2\pi)$,

$C_S: t\mapsto t\cdot e^{\frac{2\pi i}{n}} \quad (0\le t\le R)$, that is, $C=C_R * C_\ell * (-C_S)$.

Inside C, $1+z^n$ has only one zero, $\exp(i\pi/n)$, so f has only one singularity inside C, the simple pole $\exp(i\pi/n)$, by the Residue theorem,

$$\int_C f = 2\pi i\cdot \mathrm{Res}(f,\exp(i\pi/n)) = \lim_{z\to\exp(i\pi/n)}(z-\exp(i\pi/n))\cdot\frac{1}{1+z^n}\cdot 2\pi i$$

$$= \frac{2\pi i}{n\cdot\exp(i\pi/n)^{n-1}} = -\frac{\exp(i\pi/n)}{n}\cdot 2\pi i$$

Meanwhile,

$$\left|\int_{C_\ell} f\right| = \int_0^{2\pi}\left|\frac{1}{1+R^n\cdot e^{it}}\cdot\frac{Ri}{n}\cdot e^{\frac{it}{n}}\right| dt$$

$$= \int_0^{2\pi}\frac{R/n}{|1+R^n\cdot e^{it}|}\,dt \le \int_0^{2\pi}\frac{R/n}{|R^n-1|}\,dt \to 0 \text{ as } R\to\infty.$$

and

$$\int_{C_S} f = \int_0^R \frac{e^{2\pi i/n}}{1+t^n e^{2\pi i}}\,dt = e^{\frac{2\pi i}{n}}\cdot\int_0^R\frac{1}{1+t^n}\,dt. \text{ So that}$$

$$-\frac{2\pi i}{n}\cdot\exp(i\pi/n) = \lim_{R\to\infty}\int_C f = \int_0^\infty\frac{1}{1+x^n}\,dx - \int_0^\infty\frac{1}{1+x^n}\,dx\cdot e^{\frac{2\pi i}{n}}$$

$$= \left(1-e^{\frac{2\pi i}{n}}\right)\cdot\int_0^\infty\frac{1}{1+x^n}\,dx$$

$$\Rightarrow \int_0^\infty\frac{1}{1+x^n}\,dx = -\frac{2\pi i}{n}\cdot\exp(i\pi/n)\Big/\left(1-e^{\frac{2\pi i}{n}}\right).$$

▷

Evaluate $\int_0^\infty \frac{x^2}{1+x^6}\,dx$

Proof. Let C_R be a closed curve of composition $C_R = C_x * C_r * (-C_s)$ where $C_x : t \mapsto t \ (0 \le t \le R)$, $C_r : t \mapsto R \cdot e^{i\pi t/6} \ (0 \le t \le 2\pi)$, $C_s : t \mapsto te^{\pi i/3} \ (0 \le t \le R)$

Inside the curve C, $f(z) = z^2/(1+z^6)$ has only one singularity at $z_0 = \exp(\pi i/6)$. By the Residue theorem

$$\int_{C_R} f = 2\pi i \cdot \operatorname{Res}(f, z_0) = 2\pi i \cdot \lim_{z \to z_0} (z - z_0) \cdot \frac{z^2}{1+z^6} = z_0^2 \cdot \frac{2\pi i}{6 z_0^5} = \frac{2\pi i}{6 z_0^3} = \frac{\pi}{3}$$

Meanwhile, for $R > 1$,

$$\left|\int_{C_r} f\right| \le \int_0^{2\pi} \left| \frac{R^2 e^{i\pi t/3}}{1 + R^6 \cdot e^{i\pi t}} \right| \left| \frac{\pi i}{6} \cdot R \cdot e^{i\pi t/6} \right| dt$$

$$\le \int_0^{2\pi} \frac{R^3}{|R^6 - 1|} \cdot \frac{\pi}{6}\, dt \to 0 \quad \text{as } R \to \infty$$

$$\int_{C_s} f = \int_0^R \frac{x^2 \cdot e^{2\pi i/3}}{1+x^6} \cdot e^{\pi i/3} \cdot dt = -\int_0^R \frac{x^2}{1+x^6}\, dx$$

Consequently,

$$\frac{1}{3\pi} = \lim_{R \to \infty} \int_{C_R} f = 2 \cdot \int_0^\infty \frac{x^2}{1+x^6}\, dx, \text{ so } \frac{\pi}{6} \text{ is we desired.} \quad \square$$

Evaluate $\int_0^\infty \sin(x^2)\,dx$. Is it equals to $\int_0^\infty \cos(x^2)\,dx$?

Proof. Let $f(z)=e^{-z^2}$

Construct a curve C as a composition $C_1 * C_2 * (-C_3)$ where

$C_1: t\mapsto t \quad (0\le t\le R)$, $C_2: t\mapsto Re^{it} \quad (0\le t\le \frac{\pi}{4})$, $C_3: t\mapsto t\cdot e^{i\cdot\frac{\pi}{4}} \quad (0\le t\le R)$

Since f is entire, $\int_C f = 0$, by the Cauchy-Goursat's theorem.

Meanwhile,

$$\left|\int_{C_2} f\right| \le \int_0^{\frac{\pi}{4}} \left| e^{-R^2\cdot e^{2it}} \cdot Ri\cdot e^{it}\right| dt = \int_0^{\frac{\pi}{4}} e^{-R^2\cdot\cos 2t}\cdot R\,dt$$

let $u=\pi/4 - t$, then $\cos 2t = \cos(\frac{\pi}{2}-2u) = \sin 2u$

Then, $\int_0^{\frac{\pi}{4}} e^{-R^2\cos 2t}\cdot R\,dt = \int_0^{\frac{\pi}{4}} e^{-R^2\cdot\sin 2u}\cdot R\,du$

$$\le R\cdot\int_0^{\frac{\pi}{4}} e^{-R^2\cdot 4u/\pi}\,du$$

$$= R\cdot\frac{\pi}{4R^2}\cdot\left(1-e^{-R^2}\right) \to 0 \quad \text{as} \quad R\to\infty.$$

Therefore, $\lim_{R\to\infty}\int_{C_2} f = 0$. And,

$$\int_{C_3} f = \int_0^R e^{-t^2}\cdot e^{i\pi/2}\cdot e^{i\cdot\frac{\pi}{4}}\,dt = \int_0^R e^{-it^2}\cdot e^{i\pi/4}\,dt$$

$$= e^{i\pi/4}\cdot\int_0^R \cos(t^2) - i\cdot\sin(t^2)\,dt$$

$$= \frac{\sqrt{2}}{2}\cdot(1+i)\cdot\int_0^R \cos(t^2) - i\cdot\sin(t^2)\,dt$$

$$= \frac{\sqrt{2}}{2}\cdot\left(\int_0^R \cos(t^2)+\sin(t^2)\,dt + i\cdot\int_0^R \cos(t^2)-\sin(t^2)\,dt\right)$$

Now,

$$\lim_{R\to\infty}\int_C f = \int_0^\infty e^{-x^2}\,dx + 0 - \frac{\sqrt{2}}{2}\cdot\left(\int_0^\infty \cos t^2+\sin t^2\,dt + i\cdot\int_0^\infty \cos t^2-\sin t^2\,dt\right) = 0$$

So that the imagerary part is zero, thus $\int_0^\infty \cos(t^2)-\sin(t^2)\,dt = 0$, and

$$\int_0^\infty \cos(x^2)\,dx = \frac{1}{\sqrt{2}}\cdot\frac{\sqrt{\pi}}{2} = \frac{\sqrt{2\pi}}{4}. \quad \blacksquare$$

[Evaluate $\int_0^\infty \frac{\sin x}{x}\,dx$ using indented semicircle.

Proof. Let $f(z)=(e^{iz}-1)/z$, and let C be a contour of composition

$C_1: t\mapsto \varepsilon\cdot e^{it}$ $(0\le t\le \pi)$, $C_2: t\mapsto t$ $(\varepsilon\le t\le \beta)$,

$C_3: t\mapsto \beta e^{it}$ $(0\le t\le \pi)$, $C_4: t\mapsto t$ $(-\beta\le t\le -\varepsilon)$, $C=(-C_1)*C_2*C_3*C_4$.

By the Cauchy-Goursat theorem, $\int_C f(z)\,dz=0$,

Meanwhile,

$$\int_{C_1} f=\int_0^\pi \frac{e^{i\varepsilon e^{it}}-1}{\varepsilon\cdot e^{it}}\cdot\varepsilon\cdot i\cdot e^{it}\,dt=\int_0^\pi e^{-\varepsilon\cdot(\sin t-i\cos t)}-1\,dt$$

By dominant convergence theorem,

$$\lim_{\varepsilon\to 0}\int_{C_1} f=\int_0^\pi \lim_{\varepsilon\to 0} e^{-\varepsilon(\sin t-i\cos t)}-1\,dt=0$$

Evaluate $\int_0^\infty \frac{\log x}{1+x^4}\,dx$